1	**(a)**	**(i)**	gravitational force provides/is the centripetal force	B1	
			$GMm_S / x^2 = m_S v^2 / x$ (allow x or r; allow m or m_S)	M1	
			$E_K = ½ m_S v^2$ and clear algebra leading to $E_K = GMm_S / 2x$	A1	[3]
		(ii)	$E_P = – GMm_S / x$ (*sign essential*)	B1	[1]
		(iii)	$E_T = E_K + E_P$		
			$= GMm_S / 2x – GMm_S / x$	C1	
			$= – GMm_S / 2x$ (*allow ECF from* ***(a)(ii)***)	A1	[2]
	(b)	**(i)**	decreases	B1	[1]
		(ii)	decreases	B1	[1]
		(iii)	decreases	B1	[1]
		(iv)	increases	B1	[1]

(*for answers in* ***(b)*** *allow ECF from* ***(a)(iii)***)